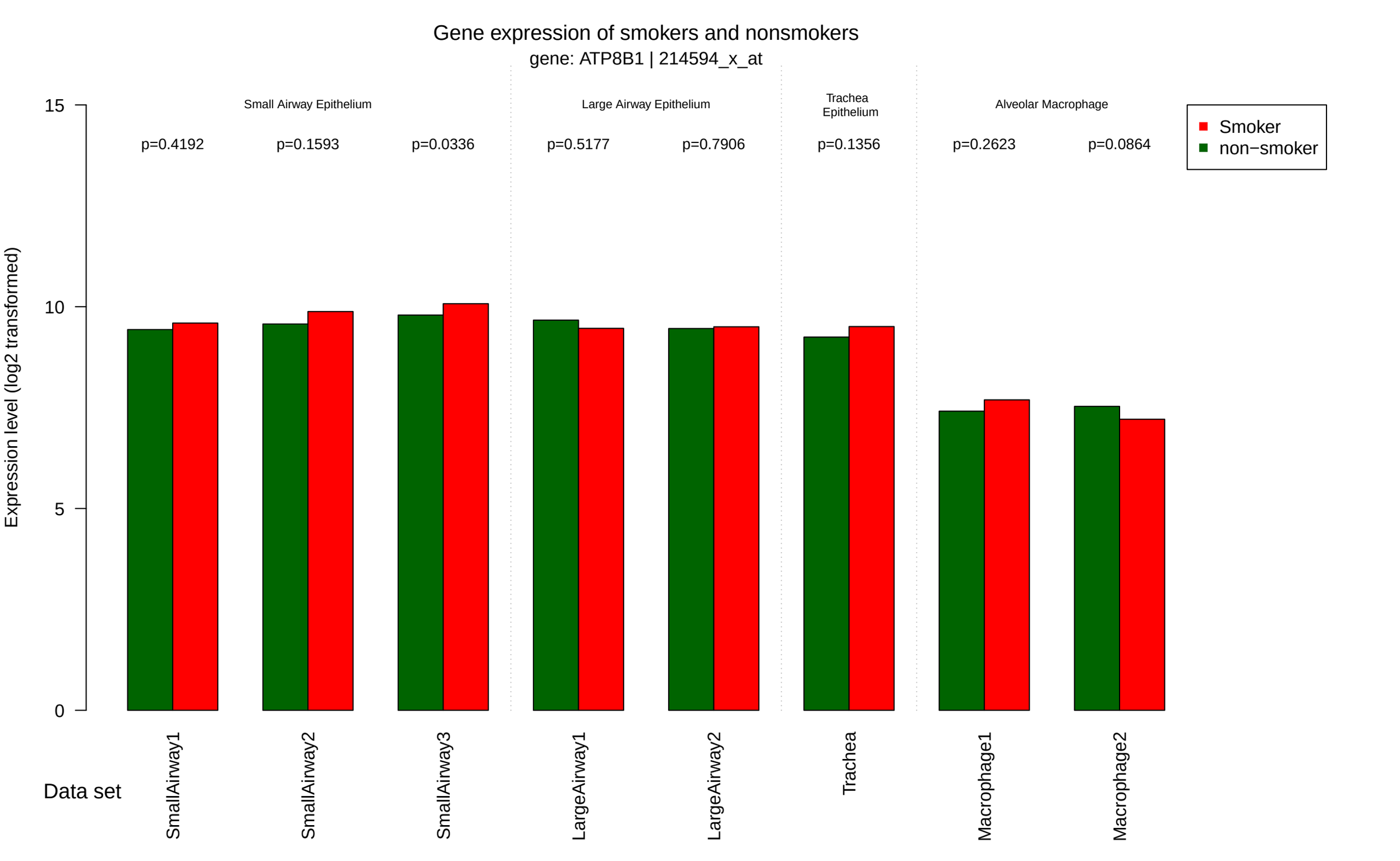
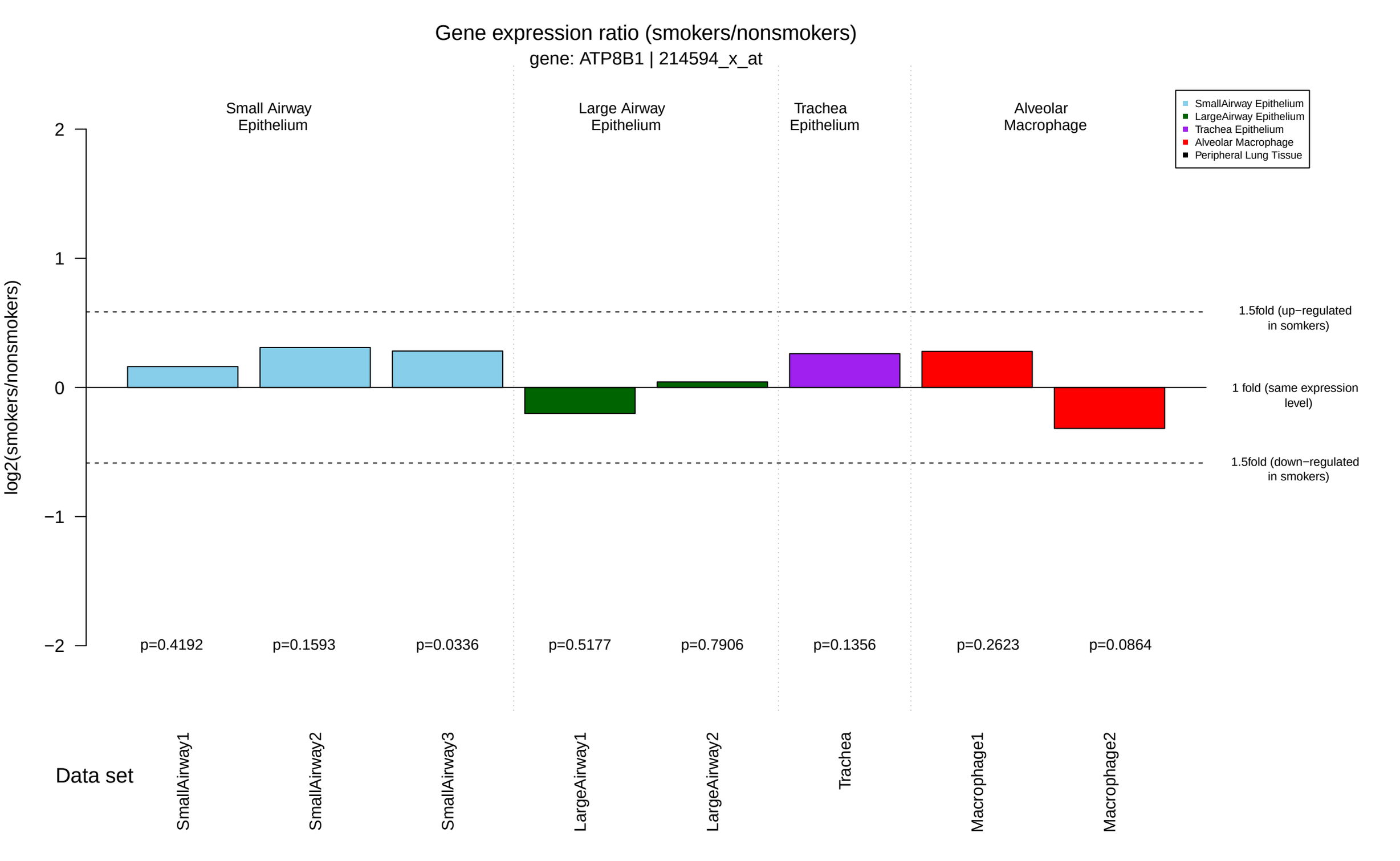
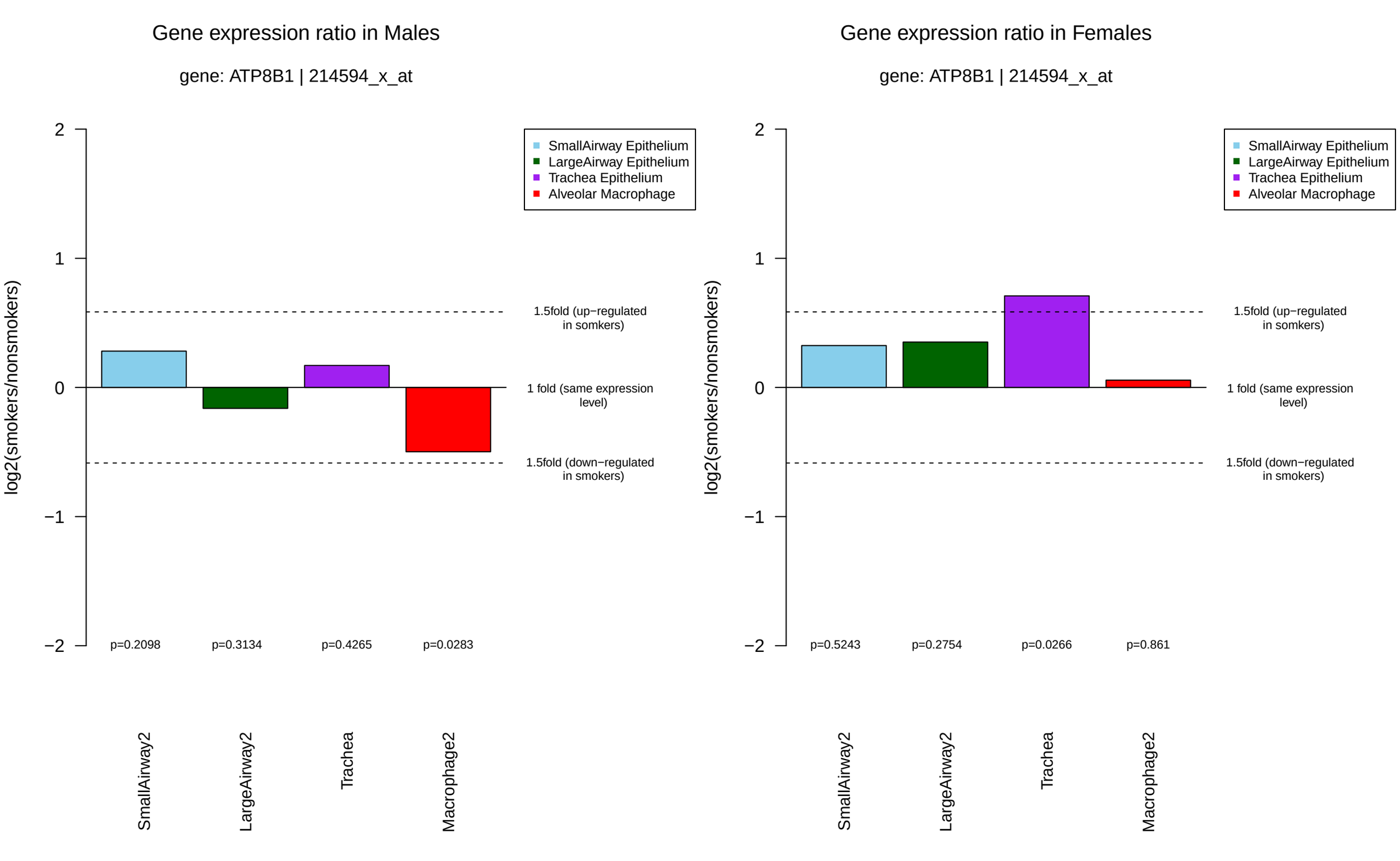
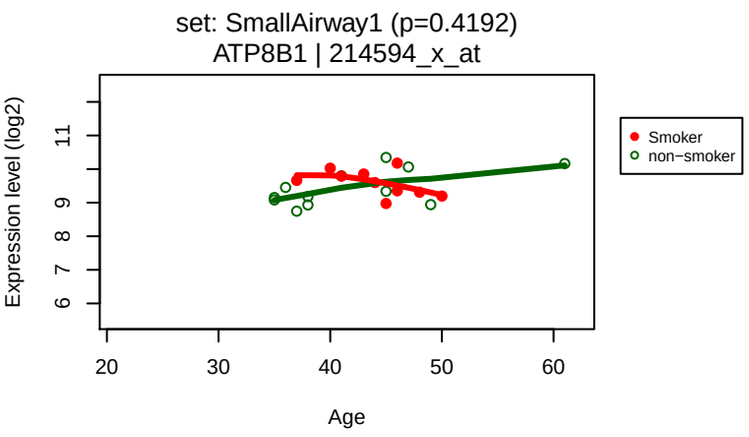
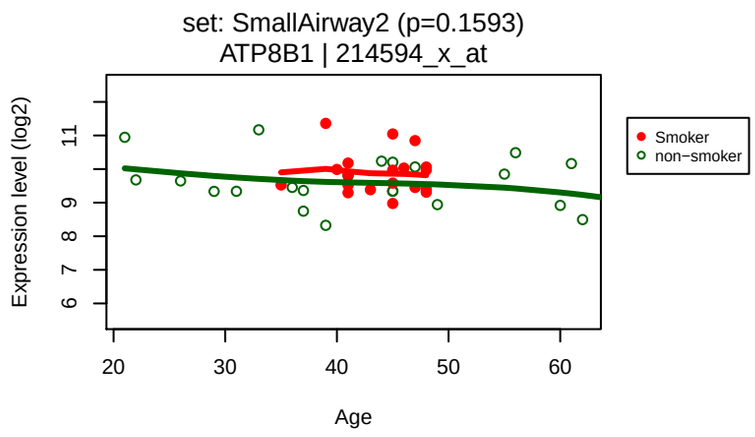
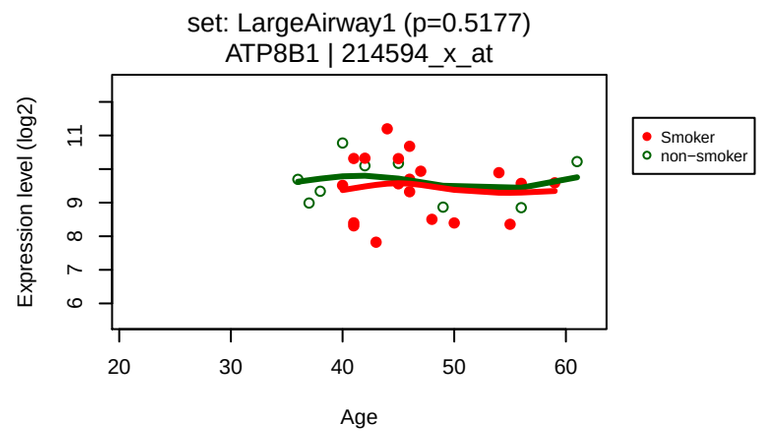

Gene expression of smokers and nonsmokers
gene: ATP8B1 | 214594_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4192
p=0.1593
p=0.0336
p=0.5177
p=0.7906
p=0.1356
p=0.2623
p=0.0864
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

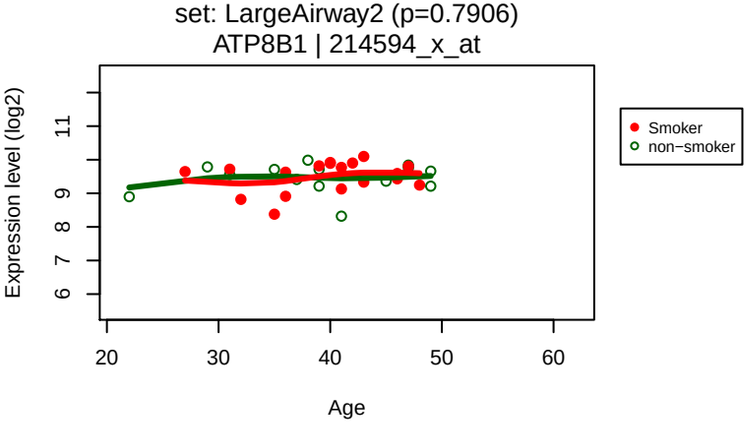

Gene expression ratio (smokers/nonsmokers)
gene: ATP8B1 | 214594_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4192
p=0.1593
p=0.0336
p=0.5177
p=0.7906
p=0.1356
p=0.2623
p=0.0864
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

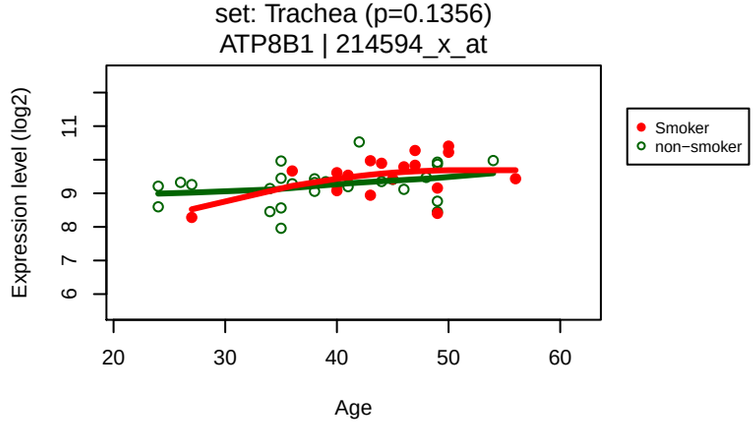

Gene expression ratio in Males
gene: ATP8B1 | 214594_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2098
p=0.3134
p=0.4265
p=0.0283
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP8B1 | 214594_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5243
p=0.2754
p=0.0266
p=0.861
SmallAirway2
LargeAirway2
Trachea
Macrophage2

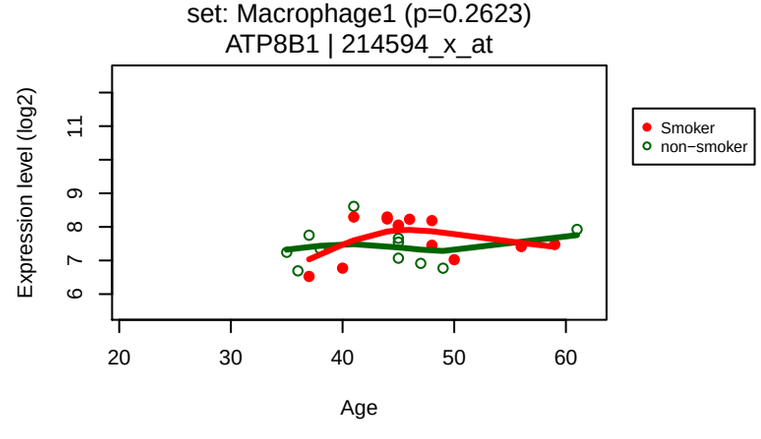

set: SmallAirway1 (p=0.4192)
ATP8B1 | 214594_x_at
Expression level (log2)
Age
Smoker
non−smoker

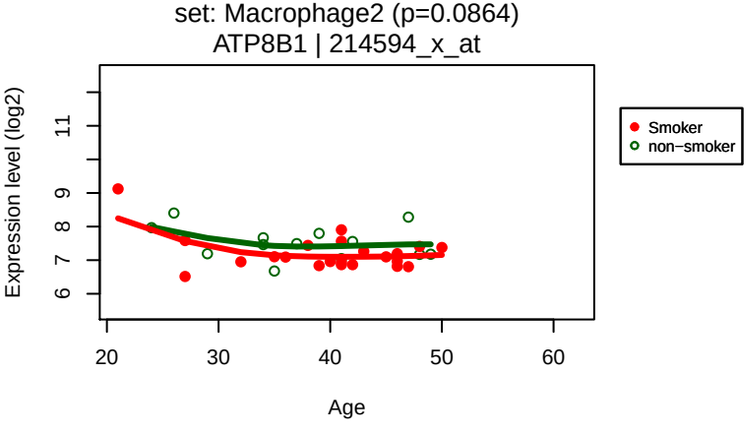

set: SmallAirway2 (p=0.1593)
ATP8B1 | 214594_x_at
Expression level (log2)
Age
Smoker
non−smoker

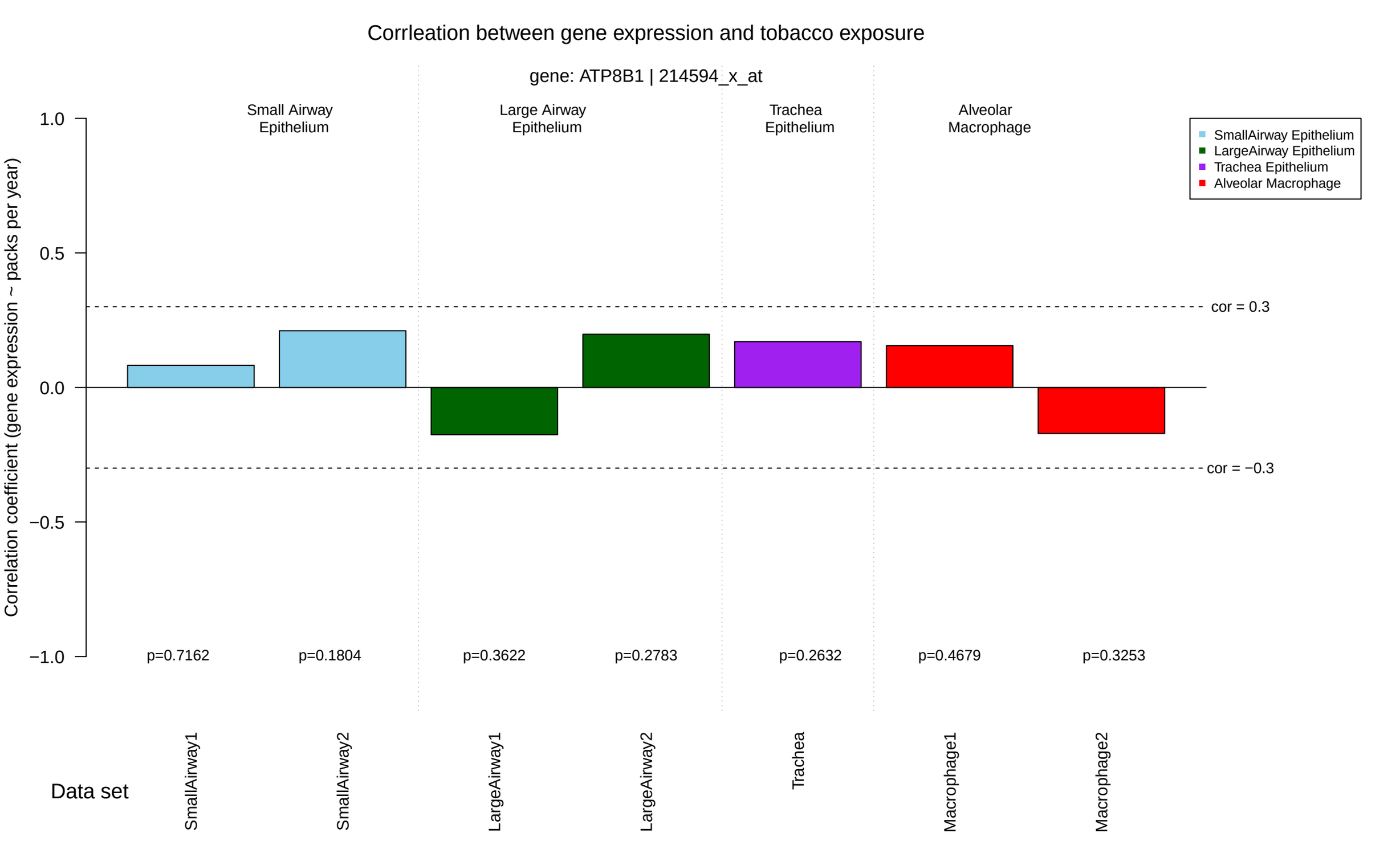

set: LargeAirway1 (p=0.5177)
ATP8B1 | 214594_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7906)
ATP8B1 | 214594_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1356)
ATP8B1 | 214594_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2623)
ATP8B1 | 214594_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0864)
ATP8B1 | 214594_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP8B1 | 214594_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7162
p=0.1804
p=0.3622
p=0.2783
p=0.2632
p=0.4679
p=0.3253
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2